Fine-Grained Action Recognition in Muay Thai Using Multimodal Large Language Models

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***Abstract*—Multimodal Large Language Models (MLLMs) are rapidly advancing artificial intelligence by enabling joint understanding across text, images, audio, and video. However, existing benchmarks often lack the fine-grained motion differentiation required for tasks such as martial arts strike recognition. This paper introduces a new benchmark dataset focused on detailed movement classification in Muay Thai and outlines an evaluation of state-of-the-art MLLM paradigms. The project investigates three model paradigms: direct video-to-text prompting, DSPy optimized prompting, and tool-using agents with pose-estimation pipelines. Through comprehensive evaluation across single-strike, combination, and multiple combination videos, this project investigates whether current MLLMs truly "understand" complex human movement or merely coarse contextual cues.**

I. Introduction

Multimodal Large Language Models (MLLMs) are rapidly advancing the capabilities of artificial intelligence (AI) by enabling joint understanding across text, images, audio, and video. Recent advancements in multimodal AI have led to increasingly powerful Vision Language Models (VLMs) capable of understanding temporal and semantic content in videos [1], [2]. Similarly, action recognition research has explored both CNN-based [4] and graph-based approaches [5], [6]; however, despite progress in general video understanding, fine-grained motion recognition remains a major challenge.

Existing VLM benchmarks focus primarily on video understanding through question and answering [3] and overlook subtle temporal variations in motion. To address this gap, this paper outlines the creation of a new benchmark dataset focused on detailed movement classification in martial arts. Martial arts, where many actions share near-identical upper body posture, stance transitions, and strike initiation, demand a degree of temporal precision and biomechanical understanding that is far beyond existing benchmarks. Current VLM datasets such as Kinetics-400 [11] provide broad action categories but do not require discriminating between highly similar movements. Even domain-specific datasets (FACTS [23], FineDiving [22]) focus on sports with clear phase segmentation or distinct tactical maneuvers, not subtle biomechanical variations within a single striking system.

There is currently no fine-grained benchmark for Muay Thai, despite its rich set of strikes, punches, kicks, knees, and elbows, that differ subtly in angle, rotation, or stance. If MLLMs are to be trusted in training assistance, coaching, or biomechanical feedback, we must first understand their limitations in this domain. This paper aims to fill the gap by introducing a curated Muay Thai dataset and evaluating multiple MLLM paradigms on tasks requiring deep temporal and structural understanding.

II. Related Work

A. Evolution of Deep Action Recognition Models

Early approaches to action recognition in video relied on traditional computer vision features and simple classifiers. With the rise of deep learning, Convolutional Neural Networks (CNNs) became the foundation for learning spatio-temporal patterns directly from raw video data. A landmark was the two-stream CNN architecture introduced by Simonyan and Zisserman [8], which processes spatial RGB frames and temporal optical flow in parallel to capture appearance and motion cues separately. Following advancements integrated spatial and temporal feature learning into one model: 3D CNNs [9] which allow networks to learn motion features. This evolution has allowed for end-to-end training on video benchmarks like UCF-101 [10] and Kinetics [11], significantly improving action classification accuracy.

In recent years, transformer-based models have been able to further improve action recognition by leveraging self-attention to capture long-range dependencies across space and time [12]–[15]. Unlike CNNs, vision transformers treat video as a sequence of image patches and model spatial-temporal relationships jointly. This approach has produced state-of-the-art results for datasets like Kinetics-400 [11]. The performance that transformers have shown in understanding fine-grained action in videos has shown that they are a powerful tool when applied.

B. Skeleton-Based Action Recognition

Another line of research focuses on skeleton-based action recognition, where the human body pose is used as input instead of raw pixels. Early pose-based methods relied on hand-crafted features or predefined body part groupings, but recent approaches use graph neural networks to learn directly from skeletal motion data. A relevant example is the Spatial-Temporal Graph Convolutional Network (ST-GCN) proposed by Yan et al. [5] which represents the human skeleton as a graph and automatically learns both spatial and temporal movement patterns from the sequence of keypoints. ST-GCN,

and the models based on it, demonstrated superior generalization on large skeleton action datasets [16], further moving the limitations of early pose-estimation-based classifiers. Skeleton-based models are the key to great performance of fine-grained action analysis when accurate pose tracking is available, as they inherently focus on the structure of human movements.

C. Vision-Language and Multimodal Models

The success of LLMs and MLLMs has recently spurred the development of vision-language models for action understanding. OpenAI has shown that aligning visual representations with natural language descriptions can produce powerful general-purpose models through CLIP [17]. ActionCLIP [18], an extension of the base CLIP model, was built for action recognition within a multimodal framework, injecting semantic context from language into video representations. This approach allowed for zero-shot and few-shot action recognition of new categories by providing textual prompts. Notably, ActionCLIP achieved top-tier performance on standard benchmarks (Kinetics-400 using a ViT backbone).

Going a step beyond image/video-text pairing models, we have seen movement toward MLLMs that integrate vision and language into one unified architecture. Deepmind's Flamingo [19] is an example of a VLM designed for few-shot learning by connecting powerful pretrained image encoders with an LLM then uses cross-modal attention to ingest video and text. Flamingo managed to achieve state-of-the-art results on various open-ended tasks like visual question answering and captioning with minimal task-specific training. Similarly, Video-LLaVA [20] extends language models to handle video by unifying video and text into one language feature space. Video-LLaVA was trained on a mixture of image and video data to align both modalities which resulted in a single LLM to reason over multi-frame visual input while delivering strong results on video understanding. These MLLMs represent a new wave of techniques and architectures where high-level semantic knowledge is coupled with visual perception, offering tools for fine-grained action recognition without relying on hand-crafted features or extensive supervised training.

D. Fine-Grained Action Recognition Datasets

Progress in fine-grained action recognition has been catalyzed by the introduction of specialized video datasets that capture subtle distinctions between similar actions. FineGym [21] is a prime example because it is a large-scale dataset consisting of gymnastic events. FineGym provides temporal segmentation of each routine into phases and sub-actions, organized in a three-level hierarchy. The level of granularity in this dataset forces models to parse the temporal structure of an activity to distinguish the subtle differences between actions. Similarly, FineDiving [22], focuses on competitive diving and contains 3000 video samples with 52 action types and 29 sub-action types with 23 difficulty degree types.

Recently, there was a growing interest in fine-grained action recognition within combat and tactical sports, which have shown their own challenges due to rapid movements and strategic nuances. For example, FACTS [23], a transformer-based model that processes raw video without using pose keypoints or wearables, has achieved state-of-the-art accuracy in recognizing fine-grained fencing and boxing maneuvers. Alongside their model, a new dataset was produced made up of 8 detailed fencing and boxing actions. Despite these advancements, fine-grained analysis of martial arts such as Muay Thai (Thai boxing) strikes has seen little attention. As of November 2025, there is no current widely used dataset focusing on Muay Thai techniques, even though this domain is rich in a variety of strikes with nuanced execution differences.

III. Methodology

This study evaluates the extent to which modern Multi-Modal Large Language Models, specifically Gemini 2.5 flash, demonstrate fine-grained understanding of Muay Thai striking actions across three levels of complexity: individual strikes, combinations, and multiple combinations. While prior work has demonstrated that MLLMs can extract coarse meaning from videos (e.g. movie scenes, OCR from images, chart interpretation) or generate commentary from sports events, few studies have examined whether these models can reliably distinguish between highly similar movements within a single sport domain. Muay Thai provides an ideal testing set due to its numerous visually similar strike types and the tactical nuance in combining movements. This methodology section describes the dataset construction, preprocessing procedures, evaluation metrics, and three distinct prompting and tool-augmented paradigms implemented during experimentation.

A. Dataset Construction and Preparation

The Muay Thai dataset used in this study was entirely self-recorded and manually annotated, enabling precise control over strike classes, recording conditions, and sequence complexity. All recordings were captured using a Samsung Galaxy S21, chosen for its stable 1080p video quality and reliable autofocus. Videos were filmed in two controlled environments: an indoor gym with a heavy bag and an open space suitable for shadowboxing. Consistent lighting and fixed tripod placement were maintained across sessions to minimize visual variation unrelated to the movements themselves.

Each raw recording session contained long continuous footage of shadowboxing, bag work, or rehearsed combinations. These longer videos were imported into DaVinci Resolve 19, where they were segmented into atomic clips corresponding to individual strikes or sequences. After segmentation, additional trimming aligned clip boundaries with the exact frame where strike initiation began and where recovery posture was achieved. Removing audio ensured that models could not rely on auditory cues such as bag impact or footwork sounds.

Annotation was performed manually, using a predefined taxonomy of 14 strike categories for individual movements and an extended list for combinations. The labeling process required frame-by-frame inspection to ensure each clip

reflected a clean, unambiguous execution of the intended strike. Ambiguous movements, such as hooks with too much vertical lift (resembling uppercuts) or knees with insufficient hip extension, were discarded to preserve dataset purity. This curation step eliminated approximately 20–25% of initially extracted clips.

The final individual-strike dataset contains 336 clips across 14 classes with 24 examples per class to maintain balance. Each clip averages 1–3 seconds, typically spanning 24–90 frames, which is enough to capture both chambering and follow-through phases of the strike. Combination sequences were recorded using both heavy-bag and shadow-boxing modalities to introduce variation in rhythm, distance management, and footwork. The 39 curated combinations emphasize commonly taught Muay Thai patterns, while the 31 multiple-combination sequences include longer strings mixing punches, kicks, knees, and stance switches.

To support DSPy optimization, the dataset was split to ensure representative samples across all classes. Unlike typical train/test splits, this partitioning was carefully tuned: DSPy received only enough examples (approximately 20%) to perform structured prompt search without allowing the model to memorize patterns. All paradigms, Baseline, DSPy, and Agent were evaluated using the same withheld test set for numerical comparability.

Importantly, no pose estimation or geometric preprocessing was applied outside the Agent pipeline. The Baseline and DSPy conditions always received raw video frames to ensure that performance reflects purely semantic and perceptual understanding rather than algorithmically extracted structure. This design isolates the role of grounding tools and establishes a fair comparison between paradigms.

B. Experimental Framework

We evaluate MLLM understanding using three paradigms, each designed to isolate different forms of reasoning:

1) **Baseline - Direct-Video Prompting:** Tests whether raw prompting and semantic priors alone can produce accurate predictions.
2) **DSPy - Prompt Optimization:** Evaluates whether structured few-shot optimization improves reasoning consistency for fine-grained distinctions [26].
3) **Agent - Agentic Pipeline with Tools:** Utilize langchain [25] and pose-estimation tools (mediapipe) [24] and explicit evidence extraction to test whether grounding improves model understanding.

All models were evaluated using quantitative metrics, including levenshtein similarity [27], F1 scores, soft accuracy groupings, and confusion matrices. Additional filtering was used to remove invalid predictions (e.g. model hallucinations producing nonexistent labels).

1) Baseline: Direct-Video Prompting: The Baseline method evaluates the simplest possible interaction with the MLLM: provide the raw video clip, a list of possible strikes with human-written natural-language descriptions, and a strict instruction to output only the predicted strike(s). This approach acts as a control for assessing MLLMs' inherent ability to use semantic priors and visual pattern recognition. Prompts follow a structured template such as Fig. 1. This prompt clarifies the number of strikes expected, the exact labels allowed, and descriptions enabling semantic grounding. This paradigm tests pattern matching, prior knowledge about human motion, and the ability to follow strict output constraints. It also explores how often MLLMs hallucinate, misclassify similar actions, or generate descriptive text despite explicit instructions, an issue observed especially in multiple combination sequences.

```
Watch this Muay Thai training video and
identify the single strike being performed.
This video contains only one strike. Return
your answer as just the strike name,
potential strikes: {unique_strikes}. Only
return the strike name, nothing else.

Available strike types with detailed
descriptions:
- jab: A quick, straight forward punch with
the lead (front) hand, arm extends
  straight forward from shoulder
- cross: A powerful straight punch with the
rear hand, arm extends straight
  forward from shoulder, body rotates
- lead_hook: A curved punch with the lead
hand, arm swings in an arc from the side
- rear_hook: A curved punch with the rear
hand, arm swings in an arc from the side
- lead_uppercut: An upward punch with the
lead hand, arm moves upward from below
- rear_uppercut: An upward punch with the
rear hand, arm moves upward from below
- roundhouse: A circular kick where the leg
swings in an arc, typically hitting with
  the shin or instep. The leg comes from the
side and wraps around; also
  called ``low kick'' when targeting the
leg.
- switch_kick: A roundhouse kick performed
after switching stance (stepping
  forward with the rear leg first)
- teep: A front push kick where the foot
pushes straight forward, like a front kick.
  The leg extends forward, not sideways.
This is different from a roundhouse.
- knee: A knee strike where the knee is
driven upward into a target
- switch_knee: A knee strike after switching
stance
- lead_elbow: An elbow strike with the lead
arm, elbow strikes forward or sideways
- rear_elbow: An elbow strike with the rear
arm, elbow strikes forward or sideways
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Fig. 1. Prompt used for the Baseline Direct-Video Prompting condition.

2) DSPy Optimization: Learning Structured Prompts and Reasoning Patterns: DSPy provides a programmatic interface for optimizing prompts, exemplars, and reasoning steps through bootstrapped few-shot learning. Rather than handcrafting improved prompts, DSPy automatically searches for better instructional and structural patterns using a small training subset (approximately 20% of the dataset).

The training set up involved two modules, depending on task complexity:

- **BootStrapFewShotWithRandomSearch** for individual strike classification
- **BootStrapFewShot** for combination and multiple combinations.

DSPy then received the same detailed strike descriptions as the Baseline, a curated subset of training examples and labels, and evaluation functions for exact-match accuracy and soft accuracy (broader groupings: punch vs. kick vs. knee vs. elbow).

The optimization goals that DSPy sought to improve upon were the stability of predictions, consistency in following strict output formats, and fine-grained discrimination between similar actions (e.g. jab vs. cross, hook vs. roundhouse). DSPy iteratively rewrites and tests candidates and because of this, it often converges on prompts that encourage multi-step reasoning, summarize distinguishing features, and constrain the output more reliably.

3) Agent Paradigm: Tool-Augmented Reasoning with Pose Estimation: The Agentic method evaluates whether MLLMs benefit from access to structured, low-level kinematic information extracted from pose-estimation pipelines. The hypothesis, inspired by VLM grounding research, is that models may misinterpret raw video but improve when given explicit anatomical coordinates or trajectories.

The Agent is a LangChain managed workflow enabling Gemini to call two external tools:

1) **MediaPipe Pose Estimator**: Extracts keypoints (e.g. wrists, elbows, knees, hips) for all frames, producing structured numerical representations of motion.
2) **Video Analyzer Tool**: Allows the agent to inspect raw video content and itegrate it with keypoint data.

A typical reasoning chain consists of analyzing the video visually first and producing an initial guess. Followed by the extraction of keypoints to confirm limb trajectories, stance, and angles. Finally, the agent reconciles the findings to produce a final classification. The goal of this agent is to mimic a human coach by observing broadly, analyzing the biomechanics, and then making a judgment.

This paradigm tests whether grounding improves fine-grained classification, how well MLLMs integrate symbolic and perceptual information, and the tool calling behavior, sequencing, and correctness.

C. Rationale for Model Paradigms and Task Design

The selection of the three paradigms, Baseline prompting, DSPy optimization, and an Agentic tool-using pipeline, was driven by the need to evaluate qualitatively different forms of multimodal reasoning. Each paradigm represents a distinct hypothesis about how modern MLLMs succeed or fail when confronted with fine-grained action understanding.

The Baseline prompting condition serves as a control for semantic prior dependence. Many VLMs are known to rely on high-level contextual cues or prototypical associations rather than precise motion analysis. In martial-arts-related videos found online (e.g., tutorials, bag work demonstrations), straight punches, round kicks, and knees often appear as canonical examples. Evaluating the Baseline condition allows us to determine how much classification ability emerges purely from semantic reasoning, prior knowledge, and coarse movement cues rather than genuine temporal modeling.

The DSPy paradigm examines whether structured prompt induction can compensate for a model's perceptual weaknesses. DSPy's few-shot optimization is especially relevant because fine-grained distinctions, such as lead versus rear variants or the difference between a hook and an uppercut, often hinge on subtle details that a model may miss without explicit reasoning scaffolds. By iteratively refining instructions, DSPy encourages the model to articulate intermediate reasoning steps, emphasize distinguishing visual attributes, and adopt a more systematic decision strategy. This paradigm tests whether improved linguistic structure can mitigate perceptual ambiguity.

The Agent paradigm addresses a complementary question: whether grounding the model in explicit skeletal or kinematic representations leads to meaningful perceptual gains. Past work on pose-based action recognition demonstrates that joint trajectories provide a strong signal for disambiguating fine-grained movements. However, it is unclear whether an LLM-guided agent can correctly integrate pose data, especially when extracted in noisy conditions. This paradigm also allows investigation into whether combining symbolic biomechanical cues with raw video analysis reduces hallucinations, stabilizes temporal reasoning, or improves segmentation of complex sequences.

Finally, the task taxonomy, individual strikes, curated combinations, and multiple combination sequences were chosen to mirror the hierarchical structure of Muay Thai movements. Individual strikes represent the atomic units of the art; combinations reflect technique chaining used in drills; and multiple-combination sequences simulate realistic training flow or sparring rhythms. By structuring evaluation across these levels, the methodology probes whether MLLMs degrade gracefully with increased temporal and structural complexity, or whether errors compound in ways that reflect fundamental perceptual limitations.

Together, these design decisions create a methodology capable of isolating the contributions of semantic priors, structured reasoning, and biomechanical grounding in multimodal understanding.

IV. Evaluation Metrics

Evaluating fine-grained video understanding requires metrics that capture both exact correctness and nuanced partial alignment between predicted and true strike sequences. Because Muay Thai strikes involve subtle biomechanical differences, such as the angle of a hook versus the linear extension of a cross, or the distinction between a teep and a knee, simple accuracy metrics are insufficient for interpreting model performance. Therefore, the evaluation in this study relies

on a combination of similarity-based, token-level, class-level, and structural measures to provide a comprehensive view of MLLM performance across individual strikes, combinations, and multiple combination sequences.

The first metric used is Levenshtein similarity, seen in Table I, which measures string-level closeness between predicted and true strike labels. This is particularly important for sequences, where a model may misclassify only one element in a multi-strike series but still demonstrate partial understanding. For example, misidentifying a "jab–cross–hook" sequence as "jab–cross–uppercut" is less severe than predicting a completely unrelated sequence such as "teep–roundhouse–elbow." Levenshtein similarity allows us to quantify these degrees of closeness numerically. This was especially valuable in the combination and multiple combination experiments, where DSPy notably outperformed both the Baseline and Agent approaches by producing predictions with consistently higher similarity scores. These results suggest that DSPy is more capable of maintaining structural coherence across sequences even when its fine-grained predictions are imperfect.

TABLE I
AVERAGE LEVENSHTEIN SIMILARITY BY METHOD

Task	Baseline	DSPy	Agent
Individual Strikes	N/A	N/A	N/A
Combinations	0.437	0.515	0.426
Multiple Combinations	0.177	0.302	0.245

To complement similarity-based evaluation, we apply F1 scores(Tables II, III, and IV), weighted according to class frequencies across the dataset. F1 provides a balanced measure of precision and recall for each strike class, making it well-suited for the inherently imbalanced nature of the dataset, certain strikes such as the jab or roundhouse occur more frequently than others such as lead uppercuts or switch knee strikes. Weighted F1 thus prevents the model from appearing to perform well simply because it overpredicts common strikes. The confusion matrices (see Appendix A 3 for full confusion matrices) generated alongside the F1 scores reveal predictable patterns of model failure. The most common confusions occur between high-similarity strikes, such as jab versus cross or hook versus uppercut, validating the intuition that visual ambiguity in human striking leads directly to linguistic ambiguity in MLLM outputs. In the combination and multiple combination settings, confusion increases dramatically as the number of possible permutations expands, illustrating that temporal reasoning remains a significant weakness for models without specialized architectural support.

A central component of the evaluation for multiple combination sequences was a Boundary Analysis 2, which examined how effectively models identified the transition points between strikes, the temporal boundaries separating one action from the next. This form of evaluation is essential because multiple strike sequences frequently involve fluid, continuous movement where the distinction between one set of techniques ending and another beginning can be subtle even for trained

TABLE II
WEIGHTED F1 SCORES BY INDIVIDUAL STRIKE

Strike Category	Baseline-F1	DSPy-F1	Agent-F1	Support
Cross	0.2857	0.4138	0.2400	18
Jab	0.3415	0.2571	0.2553	19
Knee	0.4638	0.1053	0.5246	19
lead_elbow	0	0	0	19
lead_hook	0	0.0588	0	19
lead_uppercut	0	0	0	19
low_kick	0.0714	0.0625	0	19
rear_elbow	0	0.0909	0	19
rear_hook	0	0.0769	0	19
rear_uppercut	0	0	0	19
roundhouse	0.0986	0.1639	0.0779	19
switch_kick	0.0690	0	0	19
switch_knee	0	0	0	19
teep	0.1071	0.6207	0.1642	19

TABLE III
PER-STRIKE-TYPE F1 SCORES FOR SINGLE COMBOS

Strike Type	Agent-F1	Baseline-F1	DSPy-F1	Support
cross	0.2609	0.3571	0.3478	11
jab	0.4516	0.2778	0.3500	14
knee	0	0	0	3
lead_elbow	0	0	0	0
lead_hook	0	0.2353	0.3571	8
lead_knee	0	0	0	0
lead_uppercut	0	0	0	3
lowkick	0	0	0	1
rear_elbow	0	0	0	2
rear_hook	0	0	0	1
rear_knee	0	0	0	0
rear_uppercut	0	0	0	0
roundhouse	0.1538	0	0.1765	5
switch_kick	0	0	0	3
switch_knee	0	0	0	0
teep	0	0	0	2

practitioners. The boundary results reveal that none of the three methods consistently excelled, but they exhibited distinct behavioral patterns. The Baseline method demonstrated the most stable and tightly clustered boundary accuracy, suggesting that despite its limitations in fine-grained strike identification, it tended to maintain a relatively consistent interpretation of where transitions occurred. By contrast, the Agent approach produced the widest performance range: it occasionally achieved high boundary accuracy when pose information aligned well with the underlying movement, but it also produced several extremely low scores, indicating susceptibility to misinterpreting transitional motion as either merged or fragmented strikes. DSPy fell between these two extremes, achieving moderate boundary accuracy on average but with noticeably higher variability than the Baseline. This variability reflects DSPy's tendency to enforce a consistent output structure rather than precise temporal segmentation. Overall, the Boundary Analysis highlights that while each paradigm captures some aspects of sequence structure, reliably segmenting rapid, fine-grained strike transitions remains a major challenge for current MLLMs.

A final, and unexpectedly revealing, evaluation component

TABLE IV
PER-STRIKE-TYPE F1 SCORES FOR MULTIPLE COMBOS

Strike Type	Agent-F1	Baseline-F1	DSPy-F1	Support
cross	0.2667	0.0769	0.2645	18
elbow	0	0	0	1
jab	0.4528	0.2000	0.3307	21
knee	0	0	0	1
lead_elbow	0	0	0	6
lead_hook	0	0.0000	0.1515	8
lead_roundhouse	0	0	0	1
lead_uppercut	0	0	0.0999	11
low_kick	0	0	0	0
rear_elbow	0	0.0800	0	12
rear_hook	0	0	0	0
rear_uppercut	0	0	0	1
roundhouse	0.1000	0.1667	0.1481	7
spinning_elbow	0	0	0	0
switch_kick	0	0	0	0
switch_knee	0	0	0	2
teep	0	0	0	2

was invalid prediction filtering. Because each task constrained the model to choose from a closed set of valid strike labels, any prediction falling outside this set was considered invalid and removed. High invalid-prediction rates indicate unstable reasoning, hallucination tendencies, or inability to follow strict output formatting constraints. Filtering exposed the greatest divergence between paradigms: while individual-strike Agent predictions produced only three invalid outputs, the Baseline and DSPy pipelines generated no invalid outputs. However, the Baseline and Agent paradigms generated substantial numbers of invalid labels in the combination and multiple combination experiments (28 and 40 invalid predictions for combos, and 153 and 192 invalid predictions respectively for multiple combinations). Remarkably, DSPy produced zero invalid outputs in the multiple combination setting, a result that underscores its strength in generating consistently formatted responses even under high task complexity. These filtering results, Table V, not only measure technical correctness but also serve as an important qualitative indicator of a model's compliance and stability—properties essential for any downstream coaching or analysis tool intended for real-world use.

TABLE V
TOTAL FILTERING FOR REPORT

Task	Baseline	DSPy	Agent	Total (prior filter)
Individual	0	0	3	266
Combinations	28	3	40	90
Multiple Combinations	192	0	153	244

Together, these metrics form a multilayered evaluation strategy that captures both the coarse semantic understanding and the fine-grained motor comprehension required to accurately interpret Muay Thai strikes. Levenshtein similarity reveals structural correctness in sequences, F1 scores and confusion matrices highlight class-level discriminative ability, Boundary Analysis reveals whether a model correctly interprets temporal transitions, and invalid-prediction filtering identifies stability

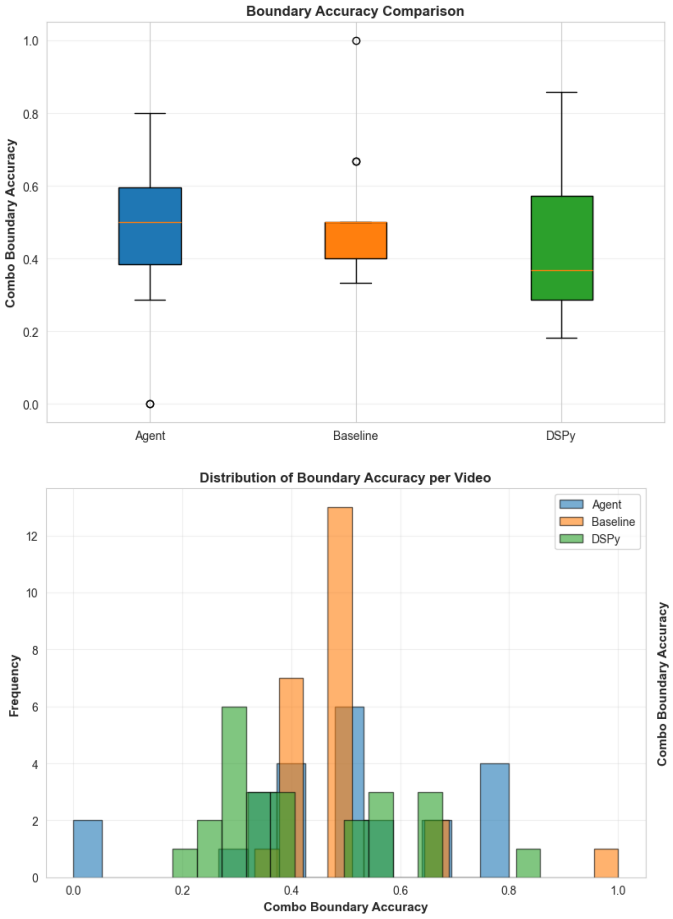


Fig. 2. Combo boundary accuracy performance for multiple-combo sequences.

in adhering to constraints. Combined, these measures illustrate that while MLLMs can extract high-level meaning from complex martial arts movements, the fine-grained biomechanical and temporal understanding required for accurate strike classification remains a significant challenge, one that DSPy mitigates some, though not perfectly, and one that the Agent method only partially addresses despite its grounding in pose data.

V. ERROR ANALYSIS

While the quantitative results presented in Section IV reveal considerable variation in performance across Baseline, DSPy, and Agent paradigms, a deeper qualitative analysis uncovers the underlying sources of error that drive these outcomes. Examining misclassifications, temporal segmentation mistakes, and characteristic failure modes provides insight into why current MLLMs, and particularly general-purpose video models, struggle with fine-grained Muay Thai strike recognition. This section synthesizes patterns observed across confusion matrices, boundary detections, and raw outputs to highlight the limitations of current multimodal reasoning.

A. Confusions Among Visually Similar Strikes

The most persistent error pattern across all three paradigms involves confusions between strikes that share similar biomechanical signatures. This is especially evident in the confusion matrices for individual strikes (3, II), where misclassifications cluster around pairs of movements with visually overlapping trajectories. For instance, lead hooks are frequently mislabeled as rear hooks or lead uppercuts, reflecting the model's difficulty in disentangling differences in elbow angle, wrist rotation, and hand path curvature. These cues require fine-grained spatial sensitivity that general-purpose vision-language models are not optimized for.

A similar trend appears with jab vs. cross confusions. Although the cross typically involves more pronounced hip rotation and originates from the rear hand, the camera angle and compactness of the motion often reduce the visual distinction to a subtle shift in shoulder alignment. When the Baseline or DSPy model underweights or misinterprets these rotational cues, the result is a systematic overprediction of the jab class. Even the Agent paradigm, despite having access to pose keypoints, fails to fully capitalize on rear-hand vs. lead-hand differentiation due to inconsistent keypoint localization during rapid punching movements.

These visually adjacent confusions reflect a broader issue: MLLMs excel at recognizing semantic categories but lack the precise geometric grounding needed to differentiate between movements that differ only by a few degrees of rotation or a slight shift in limb trajectory.

B. Failures Driven by Stance Switches and Transitional Movements

Strikes involving a stance change, such as switch kicks, switch knees, and certain multi-step combinations, proved particularly difficult. These movements require the model to interpret not only limb position but also changes in foot placement, balance distribution, and center-of-mass shifts. Failure cases typically arise when the stance switch precedes the main strike by several frames. If the model does not detect the switch early enough, it treats the strike as a non-switch variant, leading to incorrect labels in both individual-strike and combination tasks.

DSPy's constrained reasoning allowed it to handle some stance transitions more consistently than the Baseline or Agent models, but it still suffered from cases in which the preparatory step was partially occluded, too fast to capture, or visually ambiguous. The issue highlights the difficulty of reasoning over actions that rely more on footwork and weight transfer—elements that raw video models often de-emphasize relative to upper-body motion.

C. Pose Estimation Errors and Their Impact on the Agent Paradigm

The Agent model's errors are largely attributable to pose estimation inaccuracies. MediaPipe, although efficient, operates at a level of granularity insufficient for high-speed combat movements. When the limb trajectory becomes blurred during kicks or elbows, several errors occur:

- Keypoint jitter, where the elbow or knee rapidly jumps between plausible but incorrect positions.
- Missing keypoints, especially during turning motions like roundhouse kicks.
- Incorrect limb associations, where the model confuses left/right joints due to occlusion or rapid stance changes.

Because the Agent pipeline bases its reasoning on these keypoints, even minor inaccuracies lead to disproportionately large classification errors. For example, a roundhouse kick with partial occlusion may be interpreted as a teep simply because the knee and ankle momentarily align in a straight vertical plane. This creates a failure mode unique to the Agent paradigm: false confidence based on corrupted structured data.

D. Sequence-level Drift in Combination and Multiple Combination Tasks

In combination tasks, errors compound over time through a phenomenon we refer to as temporal drift. Once the model misidentifies an early strike, subsequent predictions become misaligned because the expected temporal boundaries shift. For example, if the model labels a jab–cross–hook sequence as jab–uppercut–hook, the incorrect uppercut prediction often pushes the model to expect the next strike slightly earlier or later than it actually occurs. This cascading misalignment is particularly damaging in multiple combination tasks, where sequences are longer and strike diversity is greater.

DSPy mitigated this issue somewhat due to its structured output constraints, which enforce a stable prediction format. However, even DSPy displayed drift when sequences contained repeated or near-repeated strikes (e.g., jab–cross–jab or hook–hook–roundhouse). These require the model to differentiate not only movement type but also movement timing, which remains a core challenge for MLLMs not explicitly trained for temporal segmentation.

E. Boundary Detection Patterns and Their Implications

Figure 2 shows boundary accuracy across paradigms. The Baseline model tends to under-segment sequences, merging multiple strikes into a single action when temporal gaps are small. The Agent, in contrast, tends to over-segment, interpreting momentary pauses or minor posture shifts as new strike boundaries due to keypoint noise. DSPy falls between these two extremes, but exhibits inconsistent segmentation when a sequence involves subtle rhythm changes.

These errors underscore a foundational challenge: current MLLMs lack an internal model of tempo, flow, and rhythm, elements essential for parsing martial arts actions.

VI. Limitations

Although this study introduces a new fine-grained benchmark for Muay Thai strike recognition and evaluates three modern multimodal paradigms, several limitations constrain the generality and interpretability of the results. These limitations point not only to areas of caution but also to promising directions for future work.

A. Single Subject Dataset

All videos in the dataset were recorded by a single practitioner, meaning the benchmark does not include variation in body type, limb length, stance habits, rhythm, experience level, or personal stylistic tendencies. Fine-grained martial arts movements are highly individualized, and a model trained or evaluated on a single subject may overfit to idiosyncratic motion patterns rather than genuinely learning strike-level distinctions. A multi-subject dataset would be necessary to establish generalizable performance.

B. Limited Recording Conditions

All footage was captured in controlled indoor environments with consistent lighting, background, and camera angle. Real-world applications, including coaching apps, live gym use, or sparring analysis, introduce challenges such as occlusion, camera shake, dynamic backgrounds, reflective clothing, and variable lighting. The current results reflect model performance under idealized conditions rather than realistic deployment settings.

C. Frame rate and Hardware Constraints

The Samsung Galaxy S21 recordings provide relatively clean 1080p footage, but the frame rate and compression can still fail to capture rapid transitions, especially in elbows, hooks, and switch movements. Many errors, particularly in strike segmentation, likely arise because subtle biomechanical cues occur across only 1–3 frames. Higher frame-rate recordings (e.g., 120–240 fps) would better preserve these micro-motions.

D. Pose Estimation Limitations

The Agent paradigm relies on MediaPipe, which is lightweight and fast but not designed for high-speed combat analysis. Fast strikes often cause motion blur and pose jitter, leading to inaccurate or missing keypoints. These inaccuracies propagate into the Agent pipeline and confound the intended "grounding" advantage. Stronger pose models, such as TransPose, MMPose HRNet, or graph transformer models, could significantly improve tool-augmented reasoning.

E. Dataset Size and Class Balance

While 336 individual strike clips provide a reasonable starting benchmark, several strike classes remain underrepresented relative to real-world Muay Thai usage. Rare techniques such as spinning elbows or switch knees are especially difficult for models to learn. Additionally, combination sequences represent only a modest number of curated examples, limiting the statistical power of the evaluation. A larger, more organically diverse dataset would allow deeper comparisons and more stable metrics.

F. Lack of Cross-View Generalization

All videos share the same perspective. Without alternate camera angles (front, side, 45°, overhead), models cannot learn view-invariant representations of movement. In coaching scenarios, users may hold their phone at arbitrary angles, making cross-view robustness essential.

G. Computational and Training Constraints

DSPy optimization was run on small training subsets due to cost and compute limitations. Larger bootstrapped few-shot search spaces may yield substantially stronger optimized prompts and reasoning patterns. Similarly, the Agent was not fine-tuned or trained on domain-specific data; it was evaluated in a zero-shot setting, which likely intensifies difficulties with fine-grained discrimination.

H. Evaluation Focus

The study emphasizes recognition accuracy, Levenshtein similarity, and temporal boundary detection, but does not evaluate end-user interpretability, safety, or coaching relevance. A model might misclassify a strike but still provide useful feedback, or conversely, produce confidently incorrect advice. These human-centered aspects are left for future work.

Taken together, these limitations highlight that while the benchmark provides a strong first step toward fine-grained martial arts understanding, substantial additional data diversity, improved perceptual models, and larger-scale optimization are required before such systems can be safely deployed in training contexts.

VII. Discussion & Conclusion

The results of this study demonstrate that current Multimodal Large Language Models (MLLMs), including state-of-the-art systems such as Gemini 2.5 Flash, lack the fine-grained biomechanical and temporal understanding required for reliable Muay Thai strike recognition. This finding aligns with broader trends identified in recent surveys of VLM capabilities, which note that these models perform well on coarse semantic tasks but consistently struggle with detailed motion interpretation and temporal precision [1]. Across all three paradigms evaluated in this work: Baseline prompting, DSPy-optimized prompting, and an Agentic framework incorporating a pose-estimation tool, the models exhibited substantial limitations when confronted with subtle movement distinctions, rapid strike transitions, and long multiple strike sequences.

A consistent observation is that MLLMs often capture high-level semantic information (distinguishing a punch from a kick) yet fail to differentiate visually similar strikes, such as hooks versus lead uppercuts or teeps versus roundhouse kicks. This mirrors findings in fine-grained action-recognition literature, including datasets such as FACTS [23] and FineDiving [22], which emphasize that small variations in limb angle, rotation, and timing require far more specialized perceptual modeling than current MLLMs possess. Recent benchmarks such as SCBench [30] demonstrate that current Video LLMs

perform well on high-level sports commentary tasks, yet these settings do not require the fine-grained biomechanical understanding demanded by martial arts strike recognition. The confusion matrices presented in this study reflect this exact pattern of errors, as all models tended to misclassify strikes with similar trajectories or overlapping biomechanical signatures.

Temporal reasoning posed an even more significant challenge. The Boundary Analysis revealed that none of the three paradigms consistently identified strike transition points, especially in longer or more fluid combinations. This limitation is consistent with known weaknesses in models lacking explicit temporal encoding modules [11], as well as prior findings that fine-grained segmentation requires architectures designed for long-range temporal reasoning such as 3D CNNs, ST-GCNs, or transformer-based video encoders [5], [12]. Prior work on transformer-based video architectures such as ViViT [12] and Video Swin Transformer [13] has shown that long-range temporal attention is essential for modeling fine-grained motion. TimeSformer [28], in particular, demonstrates that fully factorized space-time attention substantially improves temporal discrimination, highlighting capabilities that current MLLMs lack. The Agentic approach occasionally achieved higher boundary accuracy when pose data aligned cleanly with movement, but it also produced the widest range of failures. DSPy demonstrated more consistent formatting and structure, but did not substantially improve the perceptual understanding of transitions.

Equally important are the invalid prediction rates observed in Table V. Both the Baseline and Agent paradigms produced large numbers of hallucinated strike labels in the combination and multiple combination settings. Although DSPy avoided invalid predictions entirely in the multiple combination setting, its accuracy remained insufficient for a deployed coaching or feedback system.

These findings collectively indicate that current MLLMs lack the representational grounding necessary for fine-grained martial arts action understanding. While they can leverage semantic priors and pattern-matching heuristics, they do not yet possess the embodied reasoning or kinematic precision required to reliably parse complex human movement. In domains such as Muay Thai, where subtle biomechanical cues distinguish one strike from another, this limitation makes it impossible to safely or responsibly deploy such systems. A coaching or feedback tool built on top of these models, without addressing the core perceptual limitations illustrated in this study, would be fundamentally flawed and could put users at risk through incorrect guidance or reinforcement of improper technique.

Nevertheless, the results highlight several promising future directions. One is the integration of agentic systems with specialized deep learning tools. Rather than relying on MLLMs as primary perception engines, future architectures may instead pair them with robust video encoders, pose-estimation networks such as Mediapipe [24], or graph-based kinematic models such as ST-GCN [5]. More expressive pose encoders may also enhance agentic pipelines. Recent transformer-based pose models such as TransPose [29] provide substantially richer joint representations than lightweight systems like MediaPipe [24], suggesting a promising direction for improving kinematic grounding. In such a system, the MLLM would serve mainly as a reasoning and explanation layer operating on top of highly accurate domain-specific perceptual features. Another avenue is fine-tuning or instruction-tuning MLLMs on domain-specific datasets like the one introduced in this work, consistent with trends in sports-action research demonstrating meaningful gains through targeted adaptation [22], [23]. Hybrid pipelines of this kind align with current progress in agentic reasoning frameworks [25] and structured prompt optimization approaches such as DSPy [26].

A. Broader Implications for AI-Assisted Coaching Systems

The error patterns observed in this study have direct implications for the design of AI-based coaching tools. Human coaches rely on a nuanced interplay of timing, rhythm, balance, and intention when evaluating a strike—dimensions that current MLLMs do not model. For example, the confusion between jabs and crosses is trivial for humans because coaches prioritize rotational force, hip displacement, and weight transfer, none of which are explicitly accessible through standard visual embeddings. Likewise, stance-switch techniques carry distinctive auditory and visual cues that humans perceive intuitively, but MLLMs cannot infer. These findings suggest that any coaching application using modern MLLMs must layer additional perceptual modules, or rely on hybrid systems, to approximate human-level analysis.

B. Why Muay Thai Is an Especially Challenging Benchmark

Muay Thai presents unique difficulties not seen in other sports datasets such as FineGym or FineDiving. First, strikes often occur within a fraction of a second, meaning that decisive biomechanical cues span fewer frames than typical sampling intervals in VLM pipelines. Second, many techniques intentionally mimic others before deviating at the last moment: a feinted teep may resemble the initial chamber of a knee, and a tight rear hook can appear nearly identical to a diagonal elbow strike until the final extension. These deceptive or transitional motions, which constitute core tactical elements of Muay Thai, pose significant challenges for models not trained on combat arts or high-frequency temporal data.

Additionally, guard positions occlude key joints in a way rarely seen in other action-recognition domains. Hands obscure the jawline, elbows obscure torso rotation, and the lead shoulder frequently blocks the rear arm during punching sequences. Unlike sports with open posture (e.g., gymnastics), Muay Thai's protective mechanics inherently reduce visual clarity. This partially explains the misclassifications visible in the confusion matrices in Appendix A, where visually subtle classes collapse into one another.

C. Alignment With Human Motor Learning and Instruction

The degradation of model performance on longer sequences mirrors a well-documented phenomenon in human motor

learning: errors accumulate when learners fail to segment movement into coherent phases. The Baseline and Agent paradigms frequently merged or fragmented sequences in ways analogous to novice practitioners misunderstanding of rhythm or footwork cadence. This suggests that future MLLM-based systems may benefit from pedagogical structures modeled on human coaching, such as emphasizing phase segmentation, clear transitional markers, or simplified feedback loops, rather than raw classification.

D. Toward Future Architectures for Fine-Grained Action Understanding

The results indicate that achieving robust strike recognition will likely require architectures specifically designed for temporal coherence and embodied reasoning. Potential directions include integrating markerless motion-capture cues, using domain-specific pose transformers, or pairing video encoders with explicit biomechanical simulators. These hybrid systems could serve as perceptual "front ends," with MLLMs providing interpretability, explanation, and corrective feedback.

Ultimately, this study shows that current MLLMs are not yet capable of the fine-grained strike recognition or temporal segmentation required for safe and effective martial arts coaching. However, the methodology, dataset, and findings presented here provide a foundation for future research into hybrid agentic architectures, multimodal fine-tuning strategies, and perceptually grounded pipelines that may eventually bridge the gap between high-level semantic understanding and true biomechanical comprehension.

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APPENDIX A
FULL CONFUSION MATRICES

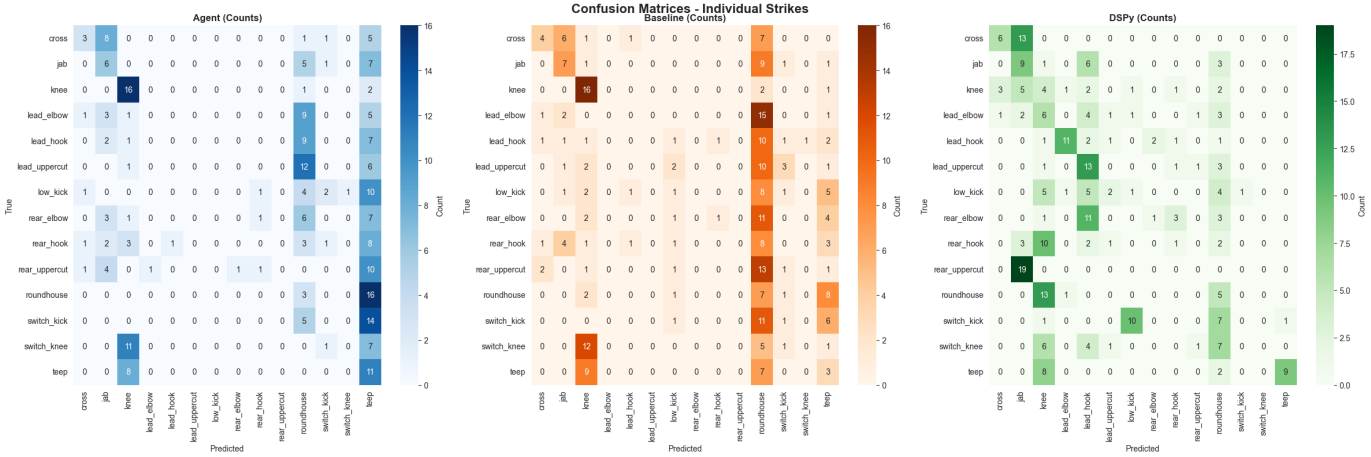


Fig. 3. Full confusion matrices for individual strikes across Agent, Baseline, and DSPy.

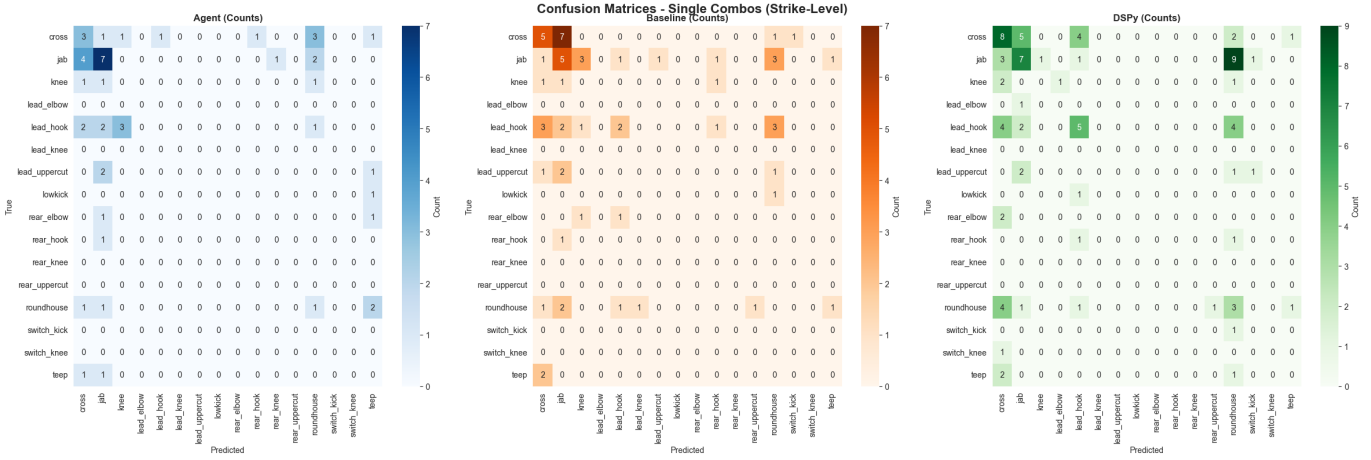


Fig. 4. Full confusion matrices for single combos at strike-level across Agent, Baseline, and DSPy.

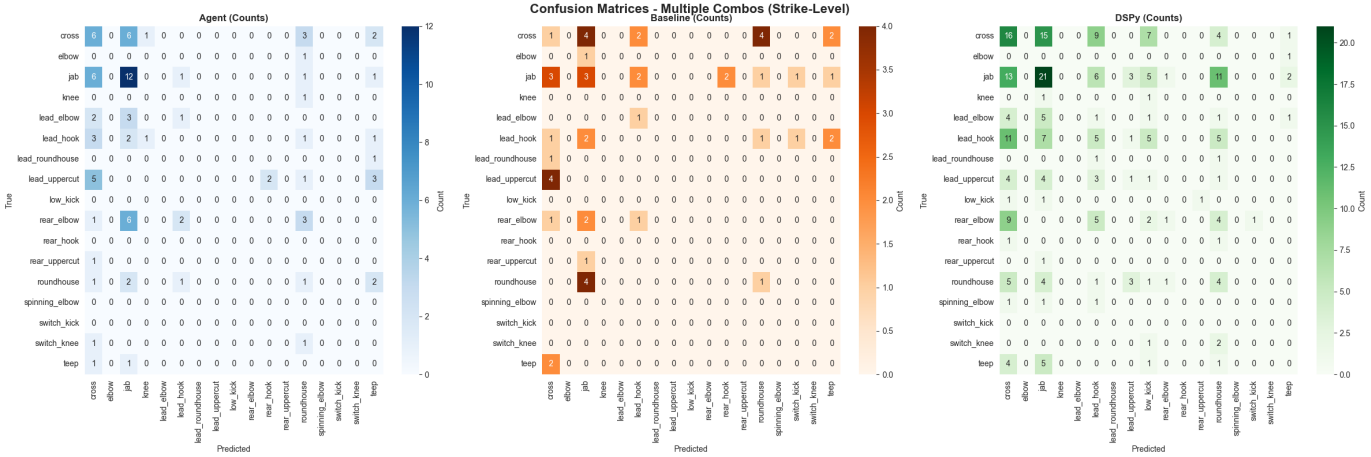


Fig. 5. Full confusion matrices for multiple combos across Agent, Baseline, and DSPy.